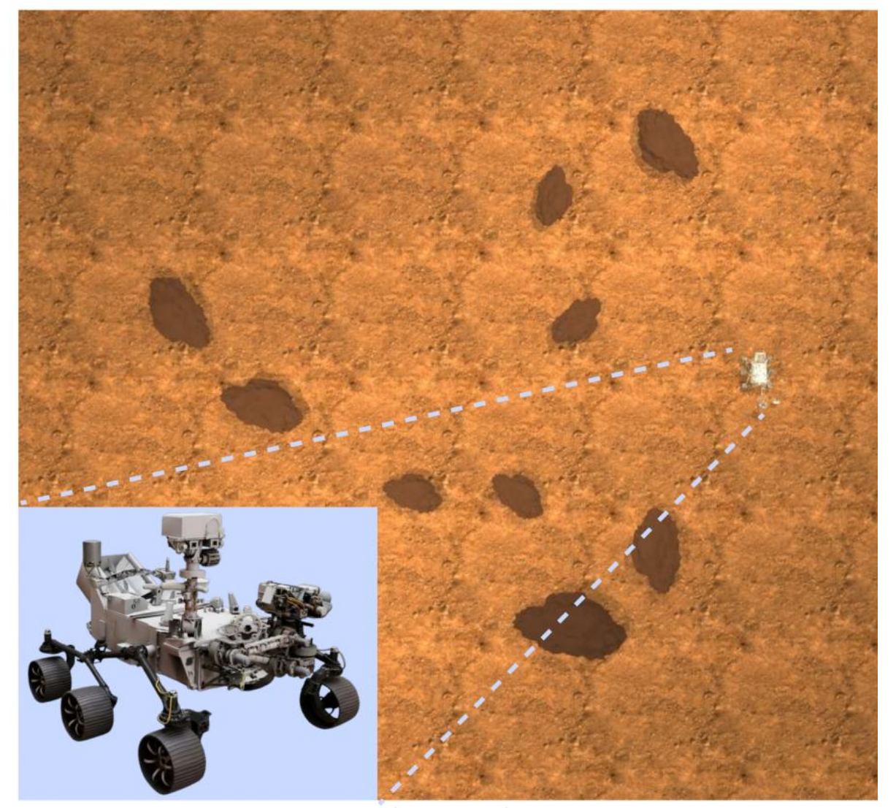
When Should a Robot Replan?
Regret-Guided Update Scheduling in Time-Varying MDPs

Negin Musavi, Gokul Puthumanaillam, Ruben Hernandez, William Schafer, Melkior Ornik
University of Illinois Urbana-Champaign
{nmusavi2, gokulp2, rubenjh2, wes6, mornik}@illinois.edu

Motivation


Operating environment

Environment
- **Stochastic motion**
  Wheels can slip on loose soil
- **Changing conditions**
  Slip rates vary

Constraint
- **Limited resources --** A strict energy budget limits sensing and onboard computation.

Goal
- Reach the destination while deciding how to allocate the available resources

? How should the agent allocate the available resources to complete the task?

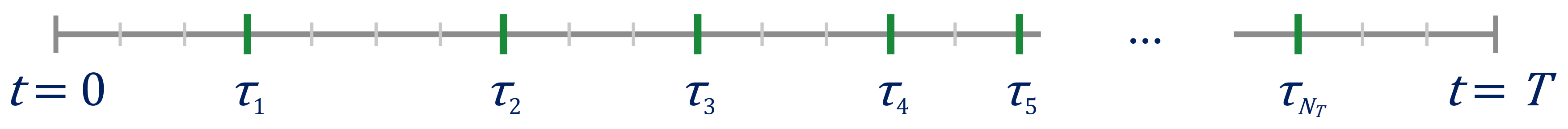
TVMDP with limited update budget

- Time-varying MDP $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, T, \{P_t\}_{t=0}^{T-1}, \{r_t\}_{t=0}^{T-1}\}$,
- $|P_t(s' \mid s, a) - P_{t-1}(s' \mid s, a)| \leq \varepsilon_{t-1}, \quad \forall s, s' \in \mathcal{S}, \forall a \in \mathcal{A},$
- Agent does not know $\{P_t\}_{t\geq 0}$ but knows $\{\varepsilon_t\}_{t\geq 0}$.
- Let $\mathcal{T}_{\text{upd}}$ be the set of time that agent is allowed to update its model and plan.
- Agent wants to maximize its cumulative reward:

$$\max_{\{\pi_t\}_{t=0}^{T-1}} \mathbb{E}\left[\sum_{t=0}^{T-1} r_t(s_t, \pi_t(s_t)) \mid s_0 = s\right] \quad \text{subject to} \quad |\mathcal{T}_{\text{upd}}| \leq B$$

Skip-update algorithm

- At selected times $\tau \in \mathcal{T}_{\text{upd}}$, the agent update its model and replan. Otherwise, act using the existing model and policy.



At $t \in (\tau_k, \tau_{k+1})$

- **Predict** -- Predict the current state via maximum-a-posteriori state estimate.
- **Act –** Apply the previously computed policy to the predicted state.

At $t = \tau_k$

- **Observe --** $\mathcal{D}_{\tau_k} = \mathcal{D}_{\tau_k - 1} \cup \{(s_{\tau_k - 1}, a_{\tau_k - 1}, s_{\tau_k})\}$
- **Estimate --** Estimate the probability transition $\hat{P}_{\tau_k - 1}$ via MLE using $\mathcal{D}_{\tau_k}$
- **Plan --** Learn a policy based on estimated transition probabilities

$$\pi_{\tau_k}^{\text{plan}} = \max_{\{\pi_h\}_{t=0}^{H-1}} \mathbb{E}\left[\sum_{h=0}^{H-1} r_{\tau_k + h}(x_h, \pi_h(x_h)) \mid x_0 = s\right] \quad \text{with} \quad x_{h+1} \sim \hat{P}_{\tau_k - 1}(\cdot \mid x_h, \pi_h(x_h))$$

Regret-guided update rule

- Evaluate the dynamic regret

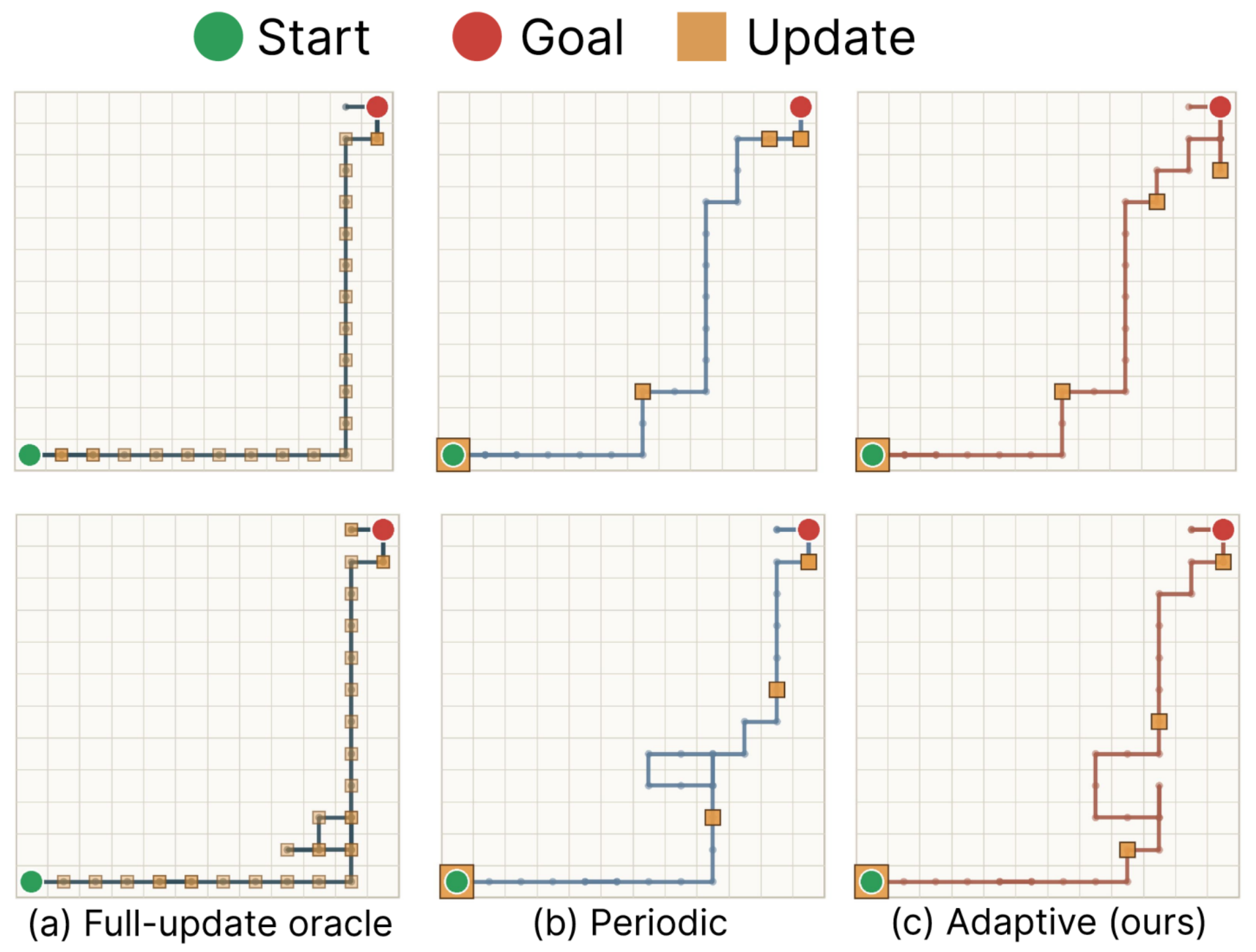
$$\mathcal{DR}(T) \leq \underbrace{\mathcal{DR}^{\text{upd}}(T)}_{\text{update-time error}} + \sum_{t \in \mathcal{T}_{\text{skip}}} \left(\underbrace{\mathcal{E}_t^{\text{hor}}}_{\text{remaining horizon}} + \underbrace{\mathcal{E}_{\tau,t}^{\text{jump}}}_{\text{immediate drift}} + \underbrace{\mathcal{E}_{\tau,t}^{\text{drift}}}_{\text{accumulated drift}} + \underbrace{\mathcal{E}_t^{\text{state}}}_{\text{state error}} \right)$$

- Combine these into a weighted score S_t to estimate the cost of continuing without an update
- Allocate updates adaptively -- Update when $S_t \geq \lambda$, where λ is a tunable threshold

Experiments

Benchmarks: 8 schedulers with the same update budget B, in two families:
- fixed-time schedules (e.g., periodic, random)
- drift-triggered rules (update when the model has changed enough).

Mars rover simulation



(a) Full-update oracle (b) Periodic (c) Adaptive (ours)

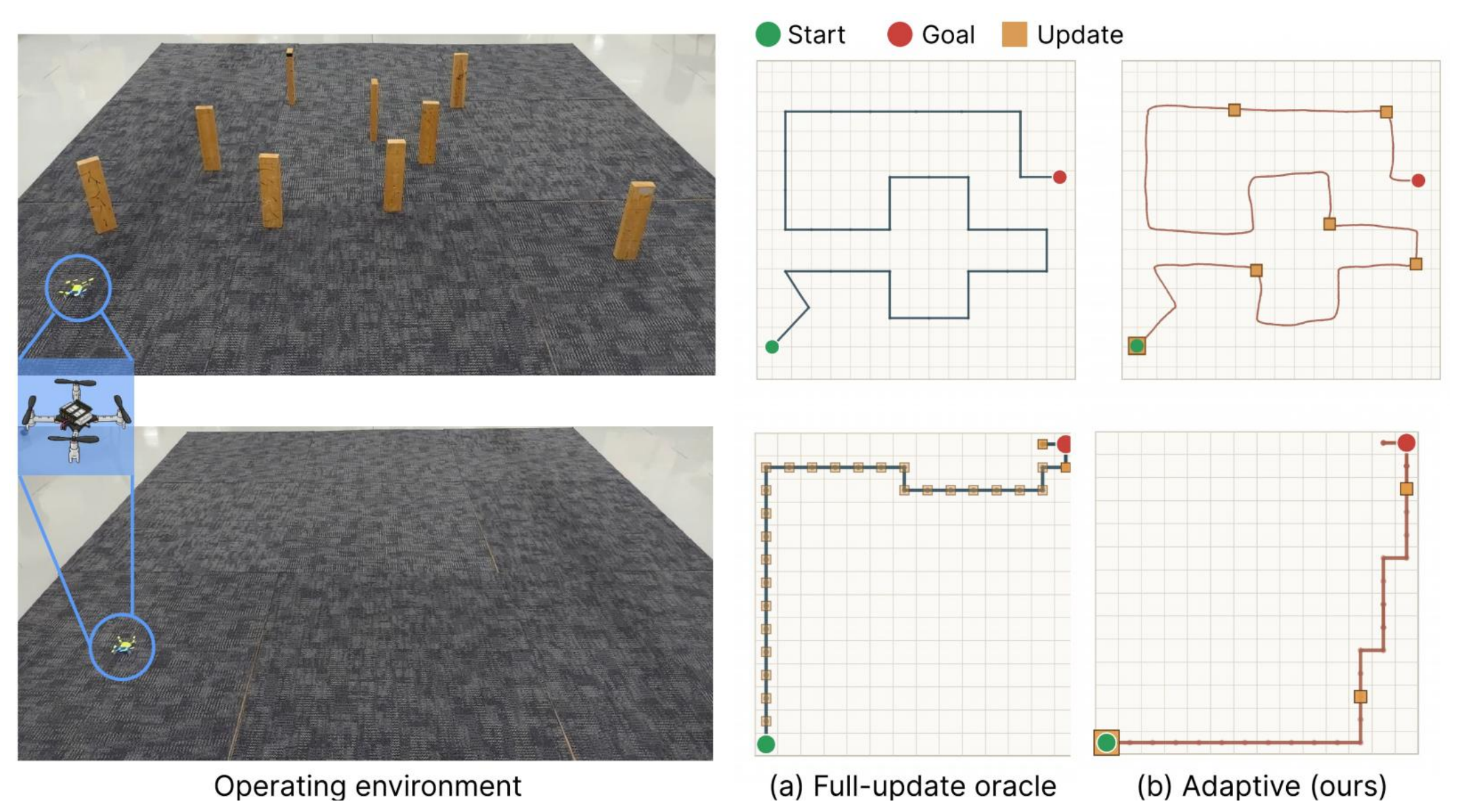
Grid navigation, 8 drift settings · 1,000 test seeds

Mean dynamic regret (lower is better)

Setting	Adaptive	Periodic	Best benchmark
Late change burst	**1.91**	4.93	2.78
Two change bursts	**0.60**	1.09	0.83
Sinusoidal change	**0.23**	0.57	0.41

- **Lowest regret in all 8 drift settings**
- **8–44% below the best benchmark** in every setting
- Largest gains when change comes in **bursts or cycles**

Crazyflie hardware



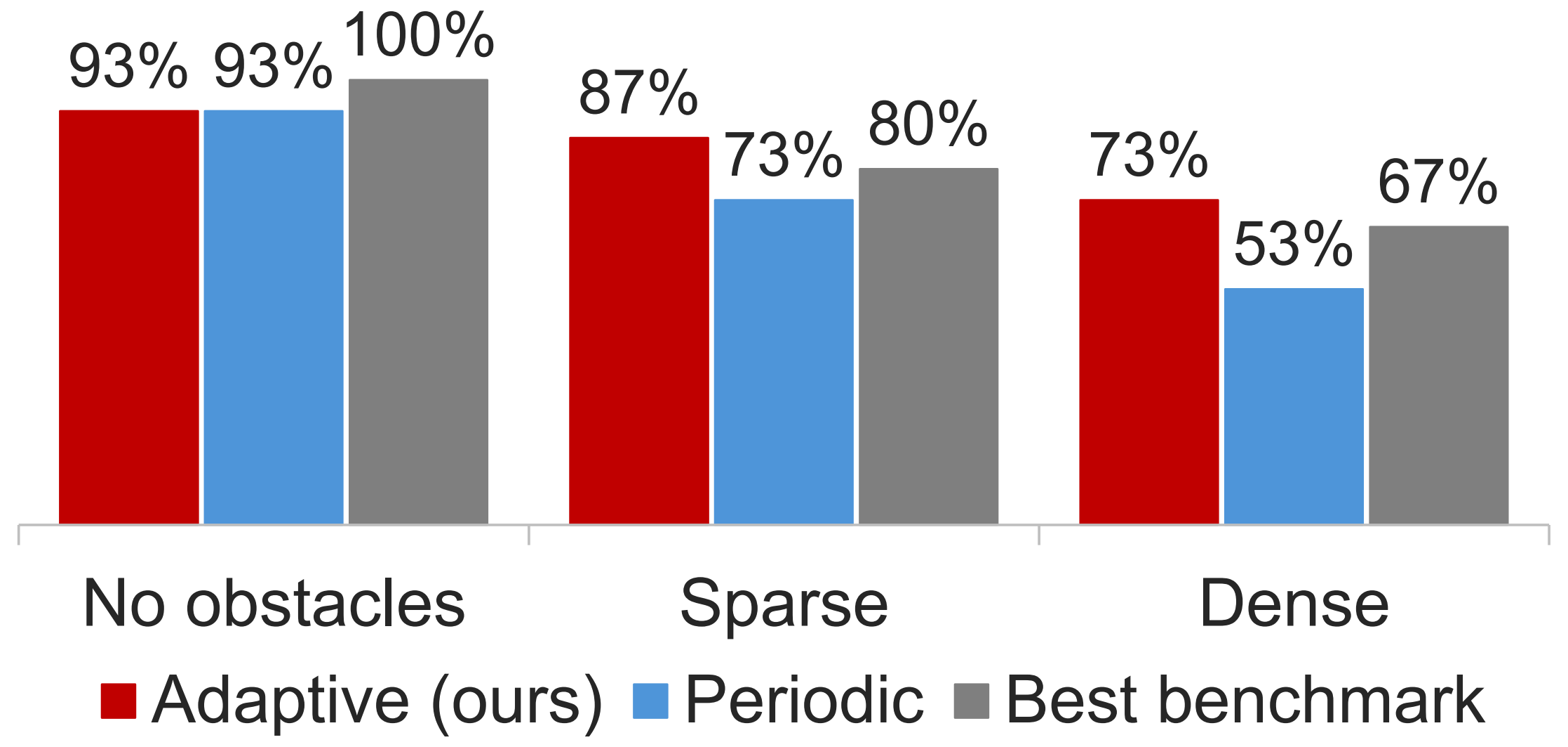
Operating environment (a) Full-update oracle (b) Adaptive (ours)

Dense obstacles

Metric	Adaptive	Periodic	Best bench.
Success ↑	**73%**	53%	67%
Collisions ↓	**0.47**	0.87	0.60
Cost ↓	**2.05**	2.70	2.20

Crazyflie quadrotor with simulated wind disturbances
15 layouts per density

Success rate by obstacle density



- **No obstacles:** every scheduler succeeds ≥ 87% — timing barely matters
- **Sparse & dense:** adaptive is best on success, collisions and cost

Next: learn the drift bounds online and extend to partially observed robots.

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